School of Mathematical Sciences | Lancaster University

Introducing basic time series models for EEG/FMRI type neural data

Rebecca Killick (r.killick@lancs.ac.uk)
June 2025

Talking Points

- Who am I?
- What is time series analysis?
- What questions can time series help you answer?
- Who, at Lancaster, can help if I want to know more?

Who am I?

- Professor in Statistics
- Specialize in collaborative impactful research
- Developing methods for analyzing data whose statistical properties change over time.

Who am I?

- Professor in Statistics
- Specialize in collaborative impactful research
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But what does that mean?

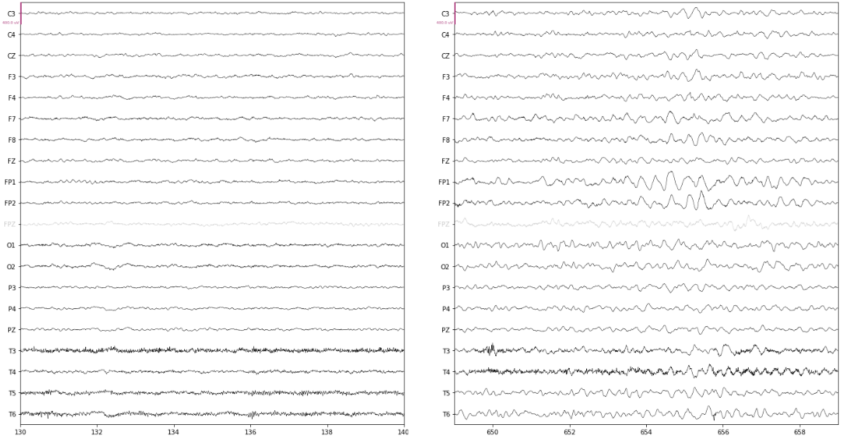
Interesting things?

. . . in collaboration

140 co-authors over 13 years.

Time Series?

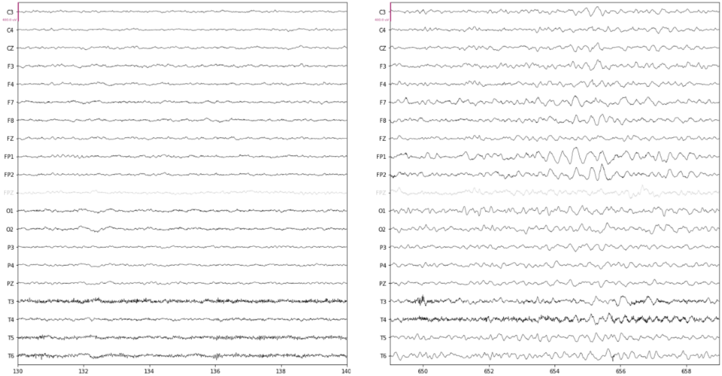
- Ordered data
- often dependent over time



(a) Background EEG (b) Rhythmic Waveforms

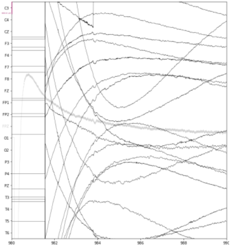
Key components

- trends
- seasonality (waves)
- dependence
- (nonstationarity)

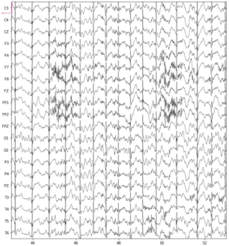


(a) Background EEG (b) Rhythmic Waveforms

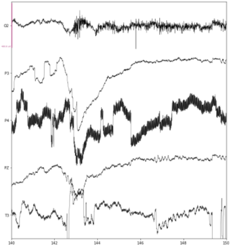
Questions - Artefacts



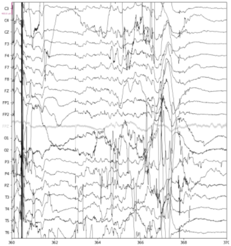
(a) Amplifier saturation found at the start of a recording.



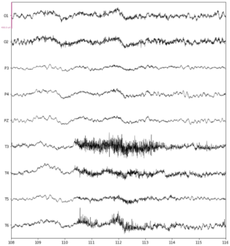
(b) Effect of ECG present in the reference electrode.



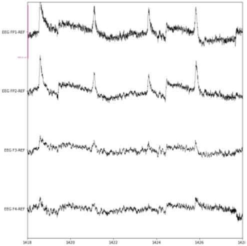
(c) Line Noise in the middle channel (P4) likely due to disconnection from the scalp.



(d) A movement artefact likely caused from full body movement.

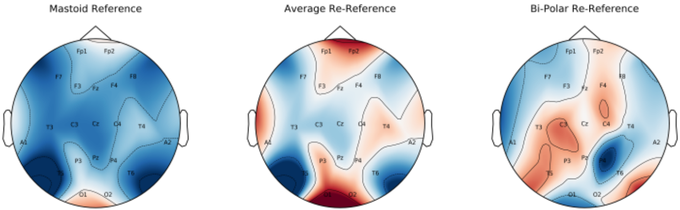


(e) An example of an artefact in the temporal electrodes caused by jaw clenching.

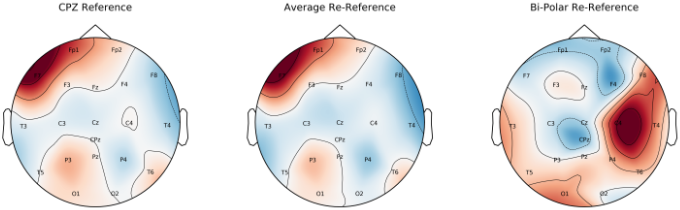


(f) Example of a patient blinking in rapid succession

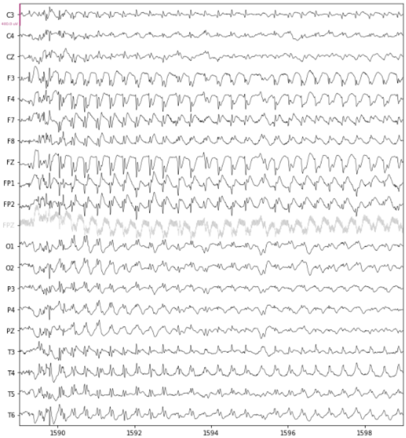
Questions- preprocessing



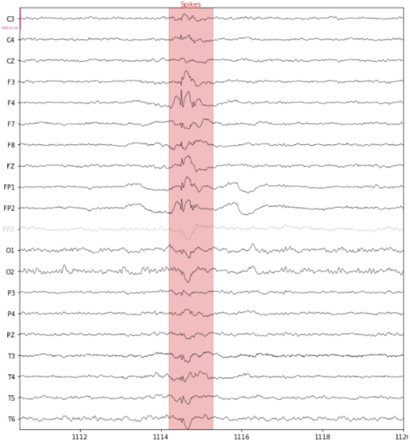
(a) Mastoid Reference



Questions - seizure

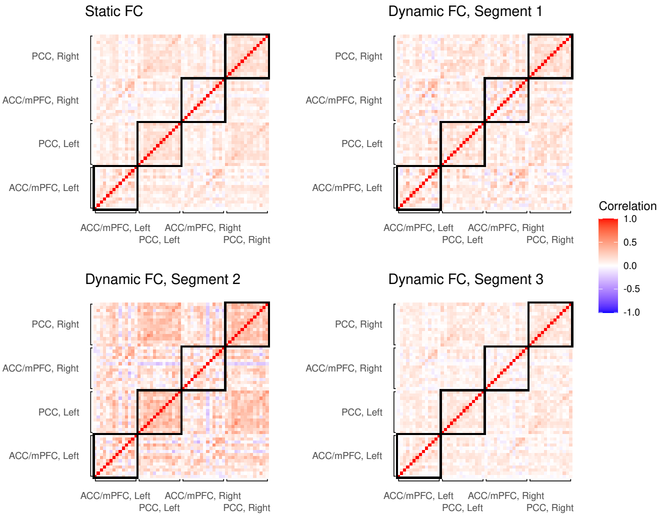


(a) Generalised Absence Seizure

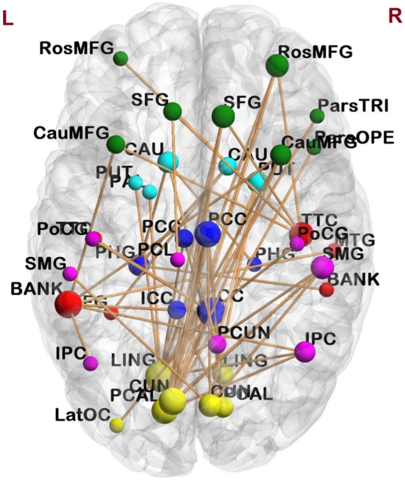


(b) Spiked Inter-Seizure Discharge

Questions - groups

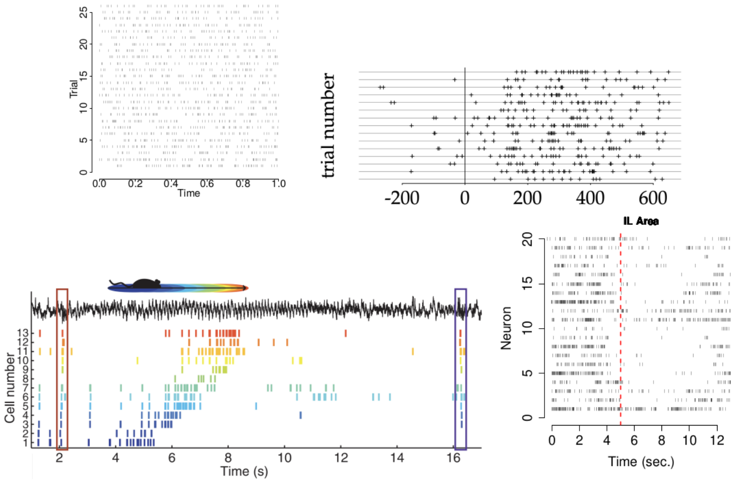


Questions - relations



https://doi.org/10.1016/j.nicl.2016.02.006

Questions - Causality



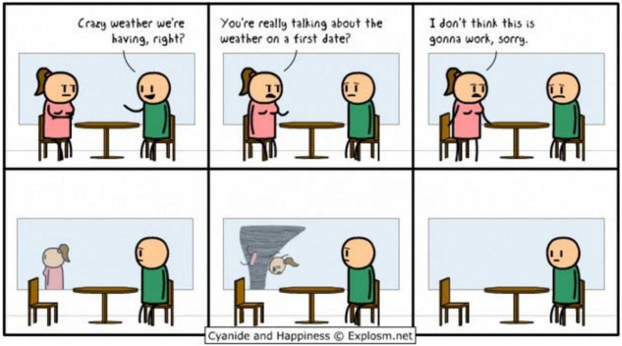
Ramezan et al. Stat. Med. (2014) www.neuroinformatics.ca

Models

- changepoint
- locally stationary Fourier/Wavelet
- multivariate
- spatiotemporal
- networks
- causal (developing)

Encouragement

Collaboration = academic dating



Who @ Lancaster?

- Myself (changes)
- Carolina Euan (spatiotemporal)
- Israel Martinez Hernandez (nonparametric)
- David Hofmeyr (clustering)
- Helen Barnett (trials and trial design)
- Florian Pein (cell biology)